Performative Prediction in a Single-Asset Market

Self-Fulfilment, Signal Erosion, and Economic Value under Forecast Adoption

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Abstract

This report studies how widespread adoption of a shared autoregressive forecasting rule affects forecast performance and economic value in a single-asset agent-based market. The market combines Brock-Hommes mean-variance demand with a Beja-Goldman / Farmer-Joshi market-maker price update. Traders may switch from a no-skill null rule to a public rolling AR forecast under either exogenous stochastic diffusion or endogenous certainty-equivalent switching. The forecast is evaluated against two distinct targets on the same path: the realised return, which is the metric a deployed forecaster actually sees, and a demand-adjusted counterfactual that strips out contemporaneous price impact.

The simulator identifies a dual-channel result. As adoption rises, the rolling out-of-sample R^2 against realised returns reaches roughly 0.19 at full adoption, against 0.07 for a zero-adoption control on the same shocks, through coordinated adopter demand. The same forecast's R^2 against the demand-adjusted return falls to roughly 0.04, against 0.08 under the control, with the rolling effective AR coefficient rising from 0.28 to 0.45 in step. Mean adopter profit is amplified rather than eroded in the cost-free baseline; the null-relative economic comparison is dominated by position-size asymmetry between the null rule and the capped adopter rule and is reported with that caveat. The headline summary is therefore that adoption raises realised predictive performance through a self-fulfilment channel while reducing the independent demand-adjusted signal the forecast was originally designed to exploit.

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3. The model

3.1 Single-asset market with linear price impact

The market is deliberately minimal. Time is discrete and indexed by $t = 0, 1, \ldots, T$. A single risky asset trades at log-price p_t, with one-period return $r_{t+1} = p_{t+1} - p_t$; the risk-free rate is normalised to zero and the asset pays no dividend, so r_{t+1} is an excess return. There are N traders, each submitting a signed order $q_{i,t}$ every period. There is no Walrasian auctioneer clearing supply against demand. Price formation instead follows the disequilibrium market-adjustment tradition, in which a risk-neutral market maker absorbs whatever aggregate order flow arrives and shifts the log-price in proportion to it [Beja and Goldman, 1980, Farmer and Joshi, 2002]. Linear price impact is the single channel through which trading feeds back into returns, and therefore the channel that carries every reflexive effect this report studies. The three equations below define the market side of the model; trader-level order formation is the subject of the next subsection.

Aggregate signed demand is the per-trader average order,

$$D_t = \frac{1}{N} \sum_{i=1}^{N} q_{i,t}, \tag{1}$$

and the market maker updates the log-price according to

$$p_{t+1} = p_t + \mu D_t + \sigma \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim \mathcal{N}(0,1), \tag{2}$$

where $\mu = 1/\lambda > 0$ is the market-maker price-impact coefficient, the reciprocal of the market maker's liquidity λ, and $\sigma > 0$ scales the exogenous news shock; equation (2) is the zero-fundamental special case of the Farmer-Joshi log-linear price update. First-differencing (2) gives a one-period market-maker return of $\mu D_t + \sigma \varepsilon_{t+1}$; the baseline return law augments it with an autoregressive component carried in from prior periods,

$$r_{t+1} = \varphi r_t + \mu D_t + \sigma \varepsilon_{t+1}, \qquad |\varphi| < 1 \text{ small}. \tag{3}$$

Equation (3) splits the return into the three pieces the rest of the model works with: a small predictable component φr_t, the signal the rolling AR rule of section 3.3 is built to estimate; the contemporaneous price impact μD_t, the only term through which trader behaviour enters; and unpredictable news $\sigma\varepsilon_{t+1}$. The φr_t term is not ad hoc: Farmer and Joshi [2002] show analytically that when part of aggregate demand comes from linear trend-following strategies, the induced return process has positive first-lag autocorrelation of the same order as the trend-following intensity, and (3) absorbs that component as a single reduced-form coefficient rather than modelling a separate trader type at baseline.

One implementation choice deserves note. In Farmer and Joshi [2002] the market maker absorbs net order flow, that is, changes in positions; the simulator's `market.py` has it absorb the level of aggregate demand each period, closer in spirit to the Beja-Goldman excess-demand adjustment, so a persistent position exerts persistent price pressure. The profit and transaction-cost accounting of later phases prices a full position turnover every period, consistent with this convention.

3.2 Trader behaviour: belief, decision, and switching layers

Trader behaviour decomposes into three layers, and keeping them separate is what later allows the results to say where each channel of the dual-channel result is produced: a belief layer (what return the agent expects), a decision layer (how that belief becomes a position), and a switching layer (whether the agent uses the shared rule at all). The mechanism under study operates on the belief layer, where the rolling AR forecast is public information, and transmits to returns only through the decision and switching layers, where action is private. At the belief layer there are exactly two states. An adopter's belief is the shared rolling autoregressive forecast $\hat{r}^{(A)}_{t+1}$ of section 3.3, identical across adopters because the rule is public. A non-adopter holds no belief about returns at all: the null rule of the same section generates order flow that is directionally uninformative on average, an active but skill-free benchmark rather than a behavioural model.

The decision layer follows Brock and Hommes [1998]: each trader is a myopic mean-variance maximiser with constant absolute risk aversion. The two equations below give the demand mapping and the capped adopter order; the order-combination rule and the profit definition appear inline in the next paragraph. With conditional forecast $\hat{r}_{i,t+1}$ of the next-period return and common perceived conditional variance $\sigma^2 > 0$, the optimal signed position is

$$z_{i,t} = \frac{\hat{r}_{i,t+1}}{a\sigma^2}, \tag{4}$$

where $a > 0$ is the coefficient of absolute risk aversion; this is the zero-dividend, zero-rate special case of the Brock-Hommes demand function. A trader buys when the expected return is positive, sells when it is negative, and submits smaller positions when the market is perceived as riskier. Adopters evaluate (4) at the public forecast rather than at an idiosyncratic belief, with the result clipped at a position cap,

$$q^{(A)}_{i,t} = \operatorname{clip}\left(\frac{\hat{r}^{(A)}_{t+1}}{a\sigma^2},\ -\bar{q}_i,\ +\bar{q}_i\right), \tag{5}$$

where the cap $\bar{q}_i > 0$ acts as a leverage or inventory constraint: rolling-window estimates of $\hat{r}^{(A)}_{t+1}$ are noisy, and the cap prevents a single extreme forecast from generating an unboundedly large trade. Non-adopters bypass the demand mapping entirely; their null order enters as drawn.

The switching layer is a single binary state: $a_{i,t} \in \{0,1\}$ indicates whether trader i has adopted the shared rule at time t. The final order combines the two rules as $q_{i,t} = (1-a_{i,t})\, q^{(0)}_{i,t} + a_{i,t}\, q^{(A)}_{i,t}$, and the one-period profit that feeds both the economic endpoint and the performance-based switching rule is $\Pi_{i,t+1} = q_{i,t}\, r_{t+1}$. How $a_{i,t}$ evolves, by stochastic diffusion or by certainty-equivalent comparison, is the subject of section 3.4. Two implementation details from `traders.py` are worth recording. First, only the product $a\sigma^2$ enters the demand mapping, so the simulator passes it as a single risk-scale parameter rather than fixing a and the perceived variance separately. Second, the advanced order returns zero whenever the forecast is not finite, so the warm-up periods before the first rolling window fills leave adopters effectively inactive instead of injecting undefined demand.

3.3 Null rule and rolling AR forecasting rule

Non-adopters trade by the null rule: trader i submits the order $q^{(0)}_{i,t} = \xi_{i,t}$ with $\xi_{i,t} \sim \mathcal{N}(0, \sigma_q^2)$, drawn independently across traders and time. The rule is deliberately a reference point rather than a model of real trading. It keeps every trader active every period while injecting order flow that is zero mean and directionally uninformative on average, so any systematic predictability or profitability the advanced rule generates is measured against a market in which trading happens but is not forecast-driven. One design feature matters downstream: the null order bypasses the demand mapping of the previous subsection and is subject to no position cap. It enters aggregate demand as drawn, with typical magnitude set by σ_q alone, while the advanced order (5) is clipped at $\bar{q}_i$. This position-size asymmetry between the two rules returns in section 6.6, where it dominates the null-relative economic endpoint in both directions.

Adopters share a single advanced rule: an autoregressive forecast of the next return, estimated on a rolling window of recent returns. With rolling-window length w and autoregressive order p, at time t the public model fits by least squares

$$r_{s+1} \;=\; c \;+\; \sum_{j=1}^{p} \varphi_j\, r_{s+1-j} \;+\; u_{s+1}, \tag{6}$$

using only the observations in the window $s = t-w, \ldots, t-1$, and issues the one-step-ahead out-of-sample forecast

$$\hat{r}^{(A)}_{t+1}(w,p) \;=\; \hat{c}_{t,w,p} \;+\; \sum_{j=1}^{p} \hat{\varphi}_{j,t,w,p}\, r_{t+1-j}. \tag{7}$$

Nothing estimated at time t uses information from after t, so the forecast is out of sample by construction, and rolling estimation lets the coefficients adapt when the return process changes, which under adoption it does: the rolling fit of (6) is exactly the object that later absorbs the amplified empirical autocorrelation of realised returns. The rule is built to recover the φr_t component of the return law (3), the simplest forecasting environment that should in principle work; penalised regressions, tree ensembles, and recurrent architectures are deliberately excluded. In the simulator, `forecast.py` implements the fit as ordinary least squares on a lagged design matrix with an intercept, the order parameter defaults to $p = 1$, and the first w entries of the forecast series are NaN, the warm-up that the non-finite-forecast guard of the previous subsection converts into adopter inactivity. The core design holds $p = 1$ with windows $w \in \{50, 100, 250\}$; higher orders $p \in \{2, 5, 10\}$ enter only as a robustness check in phase 6.

3.4 Adoption mechanisms

Two mechanisms govern how the adoption indicator $a_{i,t}$ evolves; both act on the switching layer of section 3.2 alone, and the simulator runs exactly one at a time. The stochastic mechanism comes first because it is easier to interpret: it asks whether wider use of the forecast is by itself sufficient to change predictability. The performance-based mechanism then studies the feedback between early success and later crowding.

3.4.1 Stochastic diffusion

Under stochastic diffusion each non-adopter discovers and adopts the public rule with a fixed probability $\pi \in (0,1)$ per period, and each adopter abandons it with an optional small exit probability δ:

$$P(a_{i,t+1} = 1 \mid a_{i,t} = 0) \;=\; \pi, \tag{8}$$

$$P(a_{i,t+1} = 0 \mid a_{i,t} = 1) \;=\; \delta. \tag{9}$$

The draws are independent across agents and periods, so the adoption share $A_t = (1/N)\sum_i a_{i,t}$ moves at an expected per-period rate of $\pi(1-A_t) - \delta A_t$, with no social learning and no profitability comparison: adoption is imposed on the market rather than generated by it, isolating the pure effect of wider use of a common signal. One implementation detail matters for every later comparison: adoption draws come from a dedicated random generator, so changing π or δ cannot advance the market shock stream, and regimes with different adoption settings run on paired shocks.

3.4.2 Certainty-equivalent performance-based switching

The economically meaningful mechanism makes entry depend on recent relative risk-adjusted success, scored in the same mean-variance terms as the decision layer. Each rule $k \in \{(0),(A)\}$ carries a one-period certainty-equivalent return,

$$\mathrm{CE}_t^{(k)} \;=\; \bar{\Pi}_t^{(k)} \;-\; \tfrac{a}{2}\,\hat{\sigma}_t^{2,(k)}, \tag{10}$$

where $\bar{\Pi}_t^{(k)}$ and $\hat{\sigma}_t^{2,(k)}$ are the rolling mean and variance of the rule's per-period profit over a window of length W_A, and a is the risk-aversion coefficient of the decision layer. The switching score is the gap $S_t = \mathrm{CE}_t^{(A)} - \mathrm{CE}_t^{(0)}$, and each non-adopter switches to the advanced rule with the logistic probability

$$P(a_{i,t+1} = 1 \mid a_{i,t} = 0) \;=\; \Lambda(\alpha + \beta\, S_t), \qquad \Lambda(z) = \frac{1}{1+e^{-z}}, \tag{11}$$

where α is a baseline adoption term and $\beta > 0$ controls sensitivity to recent performance; the specification is the binary Brock and Hommes [1998] discrete-choice mechanism applied at the switching layer. Two implementation choices complete the rule. First, the design specifies an entry probability analogous to (8) but no exit analogue of (9), and `adoption.py` keeps that asymmetry: adopters retain their state, so the adoption share can only ratchet upward under this mechanism. Second, the profit streams feeding (10) belong to one representative trader per rule: the common capped adopter order, and the realised order of a single representative non-adopter, which keeps the per-trader mean and variance of null profit independent of the adoption share. Unlike stochastic diffusion, this mechanism closes a second reflexive loop: early adopter success raises $\mathrm{CE}_t^{(A)}$ and accelerates entry, and the entrants' own trading reshapes the returns from which every later certainty equivalent is computed.

3.5 Intra-period timing

The equations of the preceding subsections fix what each trader and the market maker do, but not when within a period they do it, and in an agent-based market that order of operations is not implied by the equations alone. The positive-sign price-impact term μD_t in the return law (3) is what makes the order load-bearing: if adopter demand formed on today's forecast moves today's quote, then the forecast and its evaluation target are entangled within the period, and the forecast can appear self-fulfilling rather than self-eroding unless the sequence of events is made explicit. The design resolves this by fixing the following sequence inside each period, and the simulator's single state-evolution loop, `run()` in `simulate.py`, follows it exactly.

1. Agents observe returns up to and including period t and form forecasts $\hat{r}_{t+1}^{(A)}$ per (7).
2. Adopters and non-adopters submit orders $q_{i,t}$: the advanced order (5) and the null order, combined through the adoption indicator $a_{i,t}$ as in section 3.2.
3. The market maker absorbs aggregate demand D_t and moves the quote immediately, as in the price update (2).
4. The exogenous news shock $\sigma\,\varepsilon_{t+1}$ and the autoregressive term $\varphi\, r_t$ realise.

The sequence separates what the forecast knows from what its users cause. Everything the forecast may legitimately use, returns through period t, is in hand before any order is placed; everything its own adopters do to the price, the μD_t move of step 3, is in the books before next period's news arrives in step 4. With that separation fixed, it becomes well defined to score the same forecast against two different targets on the same path, the realised return that includes the impact of step 3 or a counterfactual that strips it out, which is the subject of the next subsection.

3.6 The realised vs demand-adjusted decomposition

The timing of the previous subsection makes it well defined to score the same forecast against two targets on the same path; this subsection constructs them. For evaluation purposes the return law (3) is rewritten as a decomposition into a contemporaneous price-impact component and an ex-feedback remainder. The realised return splits as

$$r_{t+1} \;=\; \mu\, D_t \;+\; x_{t+1}, \tag{12}$$

where

$$x_{t+1} \;=\; r_{t+1} - \mu\, D_t \;=\; \varphi\, r_t \;+\; \sigma\,\varepsilon_{t+1} \tag{13}$$

is the demand-adjusted return, the part of the period-$(t+1)$ return that does not depend on contemporaneous adopter trading: the lagged-return component and the exogenous news shock. The decomposition matters because the rolling AR rule of section 3.3 is, by construction, an estimate of the AR component of the return process and therefore an estimate of x_{t+1} in expectation, while the only return a deployed forecaster can observe and score against is r_{t+1}. The report accordingly evaluates the forecast against both targets as primary endpoints with asymmetric interpretations. Realised-return R^2 scores the forecast against r_{t+1} of (12): the total performance a deployed forecaster actually sees, including whatever part of the return adopter orders coordinated by the public signal have pushed in the forecast direction through μD_t, and the reading expected to rise with adoption (the self-fulfilling channel). Demand-adjusted R^2 scores the same forecast against x_{t+1} of (13): the counterfactual that strips out the contemporaneous price impact and asks whether the independent signal the forecast was originally designed to exploit survives once self-fulfilment is removed, and the reading expected to fall (the demand-adjusted-erosion channel).

One definitional caveat does enough work in the results to be worth stating exactly. The demand-adjusted return is *not* the full regression residual of an AR(1) fit on returns. Under the data-generating process (3), the full residual of a correctly specified regression is the pure exogenous news shock, $\sigma\varepsilon_{t+1} = r_{t+1} - \varphi r_t - \mu D_t$. The quantity x_{t+1} is constructed by removing the contemporaneous adopter-demand term μD_t only, leaving the lagged-return persistence channel φr_t in place. It is best read as a counterfactual: the return path the market would have produced had adopter demand not moved the quote in the current period, so that the predictable AR(1) component still has a chance to show up in the next-period observation. The distinction is the source of the erosion channel's name. The news shock is independent of everything known at time t, so against $\sigma\varepsilon_{t+1}$ any forecast carries zero signal in expectation, whatever adoption does; x_{t+1} instead retains exactly the predictable component the forecast was built to recover, so a fall in demand-adjusted R^2 records the erosion of an independent signal that was genuinely there. That is the claim the name "demand-adjusted erosion" makes, and the reading under which the channel is tracked in section 6.3 and beyond.

4. Metrics and experimental design

4.1 Primary forecast-performance endpoints

4.1.1 Realised-return R^2

The first primary endpoint scores the shared forecast against the return a deployed forecaster can observe. It is the standard out-of-sample R^2 of the one-step forecast (7) against the realised return,

$$R^2_{\text{realised}}(t) \;=\; 1 \;-\; \frac{\sum_{s=t-W}^{t-1}\left(r_{s+1}-\hat{r}^{(A)}_{s+1}\right)^2}{\sum_{s=t-W}^{t-1}\left(r_{s+1}-\bar{r}\right)^2}, \tag{14}$$

where the sums run over the most recent W forecast-target pairs ending at period t and $\bar{r}$ is the mean realised return over that window. The evaluation window is a separate knob from the AR estimation window w of section 3.3; every reported run uses $W = 1000$ periods, the notebooks' `eval_window` parameter.

The forecasts entering (14) are strictly out of sample, each $\hat{r}^{(A)}_{s+1}$ being fitted on returns through period s only. The constant benchmark follows one implementation convention worth stating once: `rolling_oos_r2` in `metrics.py` takes the trailing window's in-sample mean of the target as the benchmark, not a real-time prevailing mean recomputed from the data available before each s. The choice slightly favours the benchmark, and with near-zero-mean returns the difference is negligible. Because the target r_{s+1} contains the contemporaneous impact term μD_s of the decomposition (12), this is the total-performance reading of section 3.6, expected to rise with adoption through the self-fulfilling channel.

4.1.2 Demand-adjusted-return R^2

The second primary endpoint scores the same forecast, on the same path and over the same trailing window, against the demand-adjusted return x_{s+1} of (13):

$$R^2_{\text{da}}(t) \;=\; 1 \;-\; \frac{\sum_{s=t-W}^{t-1}\left(x_{s+1}-\hat{r}^{(A)}_{s+1}\right)^2}{\sum_{s=t-W}^{t-1}\left(x_{s+1}-\bar{x}\right)^2}, \tag{15}$$

with $\bar{x}$ the window mean of the demand-adjusted return; the benchmark convention of section 4.1.1 applies unchanged. The same forecast can earn two different R^2 readings on the same path because the targets differ by exactly that impact term: adopter orders are aligned with the forecast, so μD_s systematically moves the realised target toward the forecast, while a rolling fit that has absorbed the resulting amplified autocorrelation over-predicts the demand-adjusted target, from which the term is removed. The two endpoints are reported on equal footing, neither being the canonical evaluation target: R^2_{realised} measures what a deployed forecaster sees, and R^2_{da} measures whether the independent signal the forecast was built to exploit survives once self-fulfilment is stripped out, the demand-adjusted-erosion channel whose thresholds section 4.4 defines.

4.2 Primary economic endpoint

The primary economic endpoint of this study is net trading value: whether the forecast still pays once trading is costly. The transaction-cost extension charges a proportional cost c per unit traded, turning the gross profit $\Pi_{i,t+1} = q_{i,t}\, r_{t+1}$ of section 3.2 into a net figure read against the null rule rather than against zero: per period,

$$\Delta\Pi_{t+1} \;=\; \left(q_t^{(A)}\, r_{t+1} \;-\; c\,|q_t^{(A)}|\right) \;-\; \left(q_t^{(0)}\, r_{t+1} \;-\; c\,E|q^{(0)}|\right), \tag{16}$$

the adopter's net profit minus the null's, the latter a representative null trader's realised gross profit less its expected cost bill, with $q_t^{(A)}$ the common adopter order of (5) and $E|q^{(0)}| = \sqrt{2/\pi}\,\sigma_q$ the null rule's mean absolute position. The rule pays where the mean of (16) is positive, beating an active but skill-free benchmark net of both rules' costs.

Two properties of the benchmark shape this reading in section 6.6. First, the null's gross profit has positive mean $\mu\sigma_q^2/N$, 2.5×10^{-4} at the phase 7 parameters: a null trader's own order moves the within-period quote, booking own-impact paper profit. Second, both legs of the comparison scale with the null's position, whose mean absolute size, about 0.80, is roughly 16 times the adopter cap $\bar{q} = 0.05$. The design nevertheless treats the null-relative view as primary, since absolute profit does not control for the cost a no-skill rule itself pays whenever $c > 0$; phase 7 reports it accordingly, with the caveat that position-size asymmetry, not forecast content, dominates the comparison in both cost regimes.

4.3 Effective AR coefficient as a market-dynamics diagnostic

The two R^2 endpoints score the forecast; the rolling effective AR coefficient $\hat{\varphi}^{\text{eff}}(t)$ instead tracks the dynamics of the market the forecast trades in. It is estimated from realised returns alone, with no reference to the forecast: `rolling_phi` in `metrics.py` computes, at each period, the lag-1 sample autocorrelation of the most recent W realised returns, with the same trailing window $W = 1000$ as the R^2 endpoints in every reported run. With no adoption it should sit at the input φ of the return law (3); section 6.1 verifies that it does, with no drift. Under adoption it should rise above the input value: adopter demand responds positively to recent returns through the shared forecast, so the impact term μD_t reinforces the lagged-return component and measured persistence grows, the mechanism behind the realised-R^2 climb of section 6.3.

It is a diagnostic rather than a third endpoint because its role is explanatory: a rolling fit that has absorbed this amplified autocorrelation systematically over-predicts the demand-adjusted target, so the rise of $\hat{\varphi}^{\text{eff}}$ is the mechanical reason the two R^2 readings diverge. Sections 6.3 and 6.4 track it against the adoption share, and the relative threshold $A^*_{\varphi,\text{rel}}$ of section 4.4 summarises its rise across the parameter grid of section 6.5.

4.4 Target-specific critical adoption shares

A summary of the form "the rule stops working at adoption share A^*" is ambiguous, because the preceding subsections give three different readings of working: the realised and demand-adjusted R^2 endpoints are expected to move in opposite directions on the same path, and the economic endpoint can decouple from both. The design therefore defines one critical adoption share per evaluation target, a family of thresholds rather than a single universal number; Table 1 collects the family. Each member is the smallest adoption share at which its target metric crosses a stated criterion. Sections 6.5 and 6.6 report where each threshold fires, across the parameter grid and the transaction-cost grid respectively, which turns the dual-channel claim into a quantitative statement about which targets erode, at what share, and where in parameter space.

Table 1: The threshold family. A is the adoption share; each threshold is the smallest A at which the stated criterion is met. Relative thresholds (subscript rel) compare a rolling metric to that metric's own mean over the run's low-adoption baseline window; their factors were fixed before the phase 6 sweep was run.

Symbol	Target	Definition
$A^*_{R^2,\text{realised}}$	realised-return R^2 (14)	Primary (absolute): rolling realised-return R^2 crosses zero from above. Typically not reached, because realised R^2 rises with adoption; the absence is itself the realised-channel finding.
$A^*_{R^2,\text{realised,rel}}$	realised-return R^2	Operational analogue: realised R^2 falls to half its low-adoption baseline. Fires only as noise crossings in cells whose baseline is near zero.
$A^*_{R^2,\text{da}}$	demand-adjusted R^2 (15)	Primary (absolute): rolling demand-adjusted R^2 falls through zero.
$A^*_{R^2,\text{da,rel}}$	demand-adjusted R^2	Operational analogue: demand-adjusted R^2 falls to half its low-adoption baseline. The headline erosion threshold of section 6.5.
$A^*_{\varphi,\text{rel}}$	effective AR coefficient (§4.3)	Relative only: rolling $\hat{\varphi}^{\text{eff}}$ rises to 1.5 times its low-adoption baseline; a growth criterion, read from below, on the market-dynamics diagnostic.
$A^*_{\text{profit,abs}}$	adopter net profit	Absolute: mean net profit $q_t^{(A)} r_{t+1} - c\,\lvert q_t^{(A)}\rvert$ crosses zero at cost level c. In the regime studied the crossing runs from below, a profitability onset rather than erosion.
$A^*_{\text{profit,rel}}$	null-relative net profit (16)	Primary economic convention: the mean of (16), adopter net profit minus the null's net profit, crosses zero at cost level c.
A^*_{vol}	realised volatility	Rolling realised volatility exceeds a fixed threshold. Part of the planned metric family; not evaluated in this study.

The relative qualifier marks an operational analogue, not a redefinition. The primary definitions are absolute zero crossings, but on the two forecast-performance targets the cost-free baseline gives them little to detect: realised R^2 rises with adoption, so $A^*_{R^2,\text{realised}}$ goes unreached, and demand-adjusted R^2 falls through zero only where price impact is strong or the baseline signal is itself near zero, while at moderate settings it falls yet stays positive. The relative thresholds make erosion detectable in that regime and were fixed before the sweep was run: a fall to half the metric's own low-adoption baseline on the two R^2 targets, a rise to 1.5 times baseline on the effective coefficient, each defined only where that baseline is positive, and each carrying the rel subscript so it is never mistaken for a zero-crossing reading. The economic target runs the other way: transaction costs hand net profit a genuine zero to cross, so section 6.6 reads both profit conventions against zero directly and no relative version is needed.

4.5 Simulation protocol, seed handling, and parameter grid

Every experiment in section 6 is a set of end-to-end simulator runs with all parameters fixed in code in the owning notebook. Table 2 collects the headline parameters: the default column gives the configuration shared by the adoption phases (4 to 7, with the exceptions the caption lists), and the range column the values crossed by the phase 6 sweep or, for the adoption rate, by the phase 4 regime comparison.

Table 2: Headline simulation parameters. Defaults are the values shared by the adoption phases (4 to 7). The pre-adoption control phases differ in three places: phases 1 and 2 use $T = 5000$ and input $\varphi = 0.10$, phase 2 uses $N = 128$, and phase 3 fits windows $w \in \{50, 100, 250\}$ offline. Phase 4's saturation diagnostic extends fast-diffusion runs to $T = 30000$, and phase 5's stochastic comparator runs the fast rate $\pi = 10^{-3}$. The two shock scales are held fixed throughout, news at $\sigma = 0.01$ and null orders at $\sigma_q = 1.0$.

Parameter	Symbol	Default	Range tested
Trader count	N	200	held at default
Periods	T	8000	held at default
Price impact	μ	0.05	$\{0.025, 0.05, 0.075, 0.10, 0.15\}$
AR coefficient (input)	φ	0.25	$\{0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$
Forecast window	w	250	$\{100, 250, 500\}$
Forecast order	p	1	$\{1, 2, 5, 10\}$
Adoption rate (stochastic)	π	3×10^{-4}	$\{0, 3 \times 10^{-4}, 10^{-3}\}$
Risk scale	$a\sigma^2$	0.001	held at default
Position cap	$\bar{q}$	0.05	held at default

Seed handling has three levels. Single-path runs fix one master seed: 42 for phase 1, 43 for phase 3, 71 for phases 4 and 5. Regimes meant to be compared share that seed: phases 4 and 5 re-instantiate the generator from seed 71 for each adoption regime, and the dedicated adoption generator of section 3.4 keeps the market shock stream identical across them, so any cross-regime difference is attributable to adopter behaviour (paired shocks). Monte Carlo exercises derive per-run seeds deterministically from a notebook-level base seed: 100 seeds from base 0 in phase 2, 100 from 1000 and 50 from 5000 for phase 4's band and saturation figures, 10 per grid cell from 2000 in phase 6, and 50 from 4000 in phase 7. The pairing extends into the sweeps. Phase 6's per-run seed is offset by the window index and the (φ, μ) cell but deliberately not by the AR order, so all four orders p run on identical shock streams; phase 7 applies its eight cost levels to the same 50 simulated paths, costs entering the profit accounting only.

The phase 6 sweep crosses five price-impact values, six input AR coefficients, three windows, and four AR orders: 360 cells at 10 seeds per cell, 3600 simulator runs. The sweep's window set differs from the design set of section 3.3: it drops the shortest window $w = 50$ and adds $w = 500$, bracketing the default $w = 250$ from both sides. Each run uses slow stochastic diffusion ($\pi = 3 \times 10^{-4}$), released only after a 1500-period low-adoption baseline window has completed, so the threshold searches of section 4.4 start from $A = 0$ against a baseline measured at zero adoption.

5. Implementation

5.1 Software architecture

The simulator is a small pure-numpy Python package, `reflexive_market`, under `src/`: six working modules plus a seven-line `__init__.py` holding only a docstring and the version string, 588 lines in total. Four design constraints, fixed in the project working agreement before any simulator code was written, shaped the package. First, the mathematics uses numpy alone: AR estimation goes through `np.linalg.lstsq`, and no pandas, scipy, statsmodels, or scikit-learn import appears anywhere in `src/`. Second, every random draw flows through a `numpy.random.Generator` passed in as an explicit argument, and no module touches the global `np.random` state, so output is fully determined by the seed. Third, package functions never print or plot; parameters, figures, and the saved `npz` summaries live in the phase notebooks, which import the package. Fourth, each module implements one small, named slice of the model, so that every equation of section 3 can be traced to code; Table 3 gives the mapping.

Table 3: Modules of `reflexive_market`, their responsibilities, and the slice of the model each implements. Equation and section numbers refer to this report; line counts are from the final checkout.

Module	Lines	Responsibility	Reference
`market.py`	30	aggregate demand, market-maker price update, return law	eqs. (1)–(3)
`traders.py`	30	null random rule, demand mapping with position cap	eqs. (4), (5); §3.3
`forecast.py`	75	rolling AR(p) fit and out-of-sample forecast	eqs. (6), (7)
`adoption.py`	47	stochastic diffusion, certainty-equivalent switching	eqs. (8)–(11)
`metrics.py`	150	MSFE, out-of-sample R^2 (realised and demand-adjusted), effective AR coefficient	§4
`simulate.py`	249	`run()`: one function running T periods end to end	§3.5 timing

`simulate.py` is deliberately the largest module. Its single entry point `run()` is the only place where state evolves, and it follows the intra-period timing fixed in section 3.5: agents form forecasts on returns up to and including period t, adopters and non-adopters submit orders, the market maker absorbs aggregate demand and moves the quote, and only then do the autoregressive term and the exogenous news shock realise. Its defaults reproduce the phase 1 baseline with null traders only, so later phases extended the same function through keyword arguments (forecast window, risk scale and position cap, adoption parameters) rather than forking the loop. The two adoption mechanisms are mutually exclusive one-step pure functions invoked inside the loop. The functions in `metrics.py` score completed runs after the fact; the target-specific critical adoption shares are computed in the phase notebooks rather than in `metrics.py` because they need the full simulator output, not a return series alone.

5.2 Phased construction

The simulator was built in seven strictly ordered phases, a staging discipline fixed in the project working agreement before any simulator code was written; phase N does not start until phase $N-1$ runs end to end, meets its recorded done-when condition, and is committed. The done-when criteria for phases 1 to 3 are unchanged since the scaffold; those for phases 4 to 7 were restated under the dual-channel framing after all seven phases had been implemented, once that framing had settled as the organising description of the headline result. The ordering enforces the design's central feasibility principle: the market must work before the forecasting rule is added, the rule must work before adoption is added, and extensions come only once the baseline produces stable, interpretable results, so that any observed reflexivity effect can be traced to a small number of identifiable ingredients. Each phase owns one notebook (`notebooks/phase_NN_<name>.ipynb`) with every parameter in a single cell at the top, and saves its figures and `npz` summaries under `results/`. All seven phases are implemented; section 6 reports their results in build order.

1. **Baseline market.** Implement the single-asset market with every trader on the random null rule and confirm the empirical lag-1 autocorrelation of returns matches the input φ (section 6.1).
2. **Benchmark validation.** Rerun the null market across many seeds and verify that the descriptive statistics are stable and the rolling effective AR coefficient shows no drift; this is the no-adoption control behind every later erosion claim (section 6.1).
3. **AR forecasting rule, offline.** Add rolling AR fitting and score the forecast against realised returns while no trader acts on it, recovering a small positive out-of-sample R^2 driven by the φr_t term (section 6.2).
4. **Forecast-based trading and stochastic adoption.** Let adopters trade the shared forecast while its use spreads by stochastic diffusion; the dual-channel result first appears here (section 6.3).
5. **Performance-based adoption.** Replace diffusion with certainty-equivalent switching and compare the two adoption mechanisms on paired shocks (section 6.4).
6. **Experiments and thresholds.** Sweep (μ, φ, w, p) and locate the relative critical adoption shares $A^*_{R^2,\mathrm{da,rel}}$ and $A^*_{\varphi,\mathrm{rel}}$, defined in section 4.4, across the grid (section 6.5).
7. **Evaluation and extension.** Summarise the findings under the dual-channel framing and add the transaction-cost extension, reading A^*_{profit} against zero and against the null benchmark (section 6.6).

The phases group into three blocks. Phases 1 to 3 establish the pre-adoption control: a market with no forecast trading and a forecast with no market feedback, each verified on its own. Phases 4 and 5 switch the feedback on and establish the dual-channel mechanism under two different adoption processes. Phases 6 and 7 quantify the mechanism, first across the parameter grid and then in economic terms, with phase 7 also serving as the evaluation summary of the whole build.

5.3 Reproducibility

Every number and figure in this report can be regenerated from the repository checkout. Three conventions from the working agreement carry the load. First, every notebook fixes its randomness in the parameters cell: single-run notebooks expose a scalar `seed` (42 for phase 1, 43 for phase 3, 71 for phases 4 and 5), and multi-seed notebooks a `base_seed` and a seed count from which each per-run seed is derived deterministically. Because every random draw flows through the explicitly passed generator (section 5.1), the same seed and the same parameters reproduce the same run, and with it the same figure; the test suite asserts this determinism directly. Second, every artifact names its origin: figures save to `results/figures/phase_NN_<name>.png` and numeric summaries to `results/data/phase_NN_<name>.npz`, so each file traces to one phase notebook. The `npz` summaries store the small arrays behind the report's tables and quoted numbers, so a reader can verify any of them without rerunning a simulation. Third, the environment is thin. The package requires Python 3.10 or newer and depends at runtime on numpy, matplotlib, and joblib alone, per the pure-numpy commitment of section 5.1; `requirements.txt` adds only jupyter and pytest, for running the notebooks and the tests.

These conventions are enforced, not aspirational. The repository ships a 43-test pytest suite covering package import and per-phase invariants for phases 1 through 5, from seed determinism and recovery of the input φ to position-cap clipping and the mutual exclusion of the two adoption mechanisms. A single GitHub Actions workflow runs this suite on every push and pull request: it installs the dependencies and the package on Python 3.11 and runs `pytest -q`, so every push is checked against the full suite.

6. Results

6.1 Baseline market and benchmark validation (phases 1 and 2)

Phase 1 checks that the simulator reproduces the baseline market of section 3.1 when every trader follows the random null rule of section 3.3. A single run uses $N = 200$ traders, $T = 5000$ periods, price-impact coefficient $\mu = 0.05$, input autoregressive coefficient $\varphi = 0.10$ in the return law $r_{t+1} = \varphi r_t + \mu D_t + \sigma \varepsilon_{t+1}$, news scale $\sigma = 0.01$, null-order scale $\sigma_q = 1.0$, and seed 42. Null orders are independent Gaussian draws, so aggregate demand D_t is directionally uninformative and feeds the return only as unpredictable price-impact noise; the only structure in returns that a forecast could exploit is the φr_t term. The run behaves accordingly. The log price wanders without systematic drift (the mean one-step return of 2.0×10^{-4} sits within 1.4 standard errors of zero), and the return series is roughly stationary with standard deviation 0.0105. The empirical lag-1 autocorrelation of returns is 0.103, matching the input $\varphi = 0.10$ within the sampling error of roughly $1/\sqrt{T} \approx 0.014$ for a series of this length.

Phase 2 reruns the null-rule market across 100 independent seeds (0 through 99) with $N = 128$, $T = 5000$, and the same market parameters, to confirm the baseline is a stable control rather than a one-seed accident. The per-seed descriptive statistics cluster tightly around their theoretical values. The per-seed mean return averages 1.5×10^{-5} with a cross-seed standard deviation of 1.6×10^{-4}, consistent with a zero unconditional mean; the per-seed return standard deviation averages 0.0110 with cross-seed spread 0.00012; the per-seed lag-1 autocorrelation averages 0.103 with spread 0.012, clustered around the input $\varphi = 0.10$; and the per-seed excess kurtosis averages -0.0065 with spread 0.071, indistinguishable from the Gaussian value of zero. The kurtosis reading, together with the symmetric single-seed return histogram from phase 1, supports treating null-rule returns as approximately Gaussian, as expected when the news shock and the individual null orders are both Gaussian and aggregate demand averages across many traders. The sharpest control statistic is the rolling effective AR coefficient shown in Figure 1. Estimated on a 1000-period sliding window and averaged across seeds, the trace sits near the input value and stays flat: its total range over the 4001 periods with a complete window is 0.016, with no trend in either direction. This subsection is the control that the rest of the results lean on. Later phases attribute movements in forecast R^2 and in the effective AR coefficient to adoption of the shared forecast; that attribution is meaningful only because the same market with no adoption, simulated over the same horizon, shows no comparable drift.

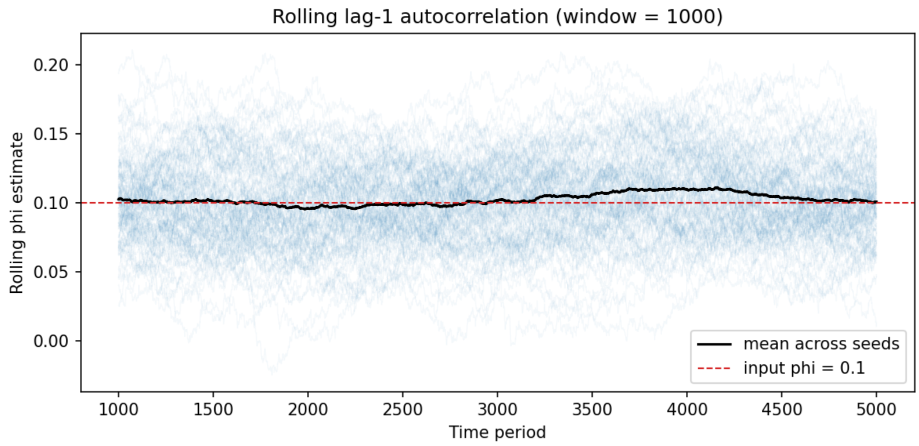

Figure 1: Rolling lag-1 autocorrelation of returns under the null rule, estimated on a 1000-period sliding window for each of 100 seeds (faint lines). The black line is the cross-seed mean and the dashed line marks the input $\varphi = 0.10$. The mean trace stays flat for the full run, with a total range of 0.016 over the 4001 periods where the window is complete, so the effective AR coefficient shows no drift in the absence of adoption.

6.2 AR forecast performance without adoption (phase 3)

Phase 3 attaches the rolling AR(1) forecasting rule of section 3.3 to the null-market design of the previous subsection, in offline mode: the forecast is estimated and scored, but no trader acts on it. A single run uses $N = 200$ traders, $T = 8000$ periods, $\mu = 0.05$, $\sigma = 0.01$, $\sigma_q = 1.0$, and seed 43, with the input autoregressive coefficient raised to $\varphi = 0.25$ so the rule has a clearly detectable signal to recover. Each period the rule fits an AR(1) by least squares on the most recent $w \in \{50, 100, 250\}$ returns and issues a strictly out-of-sample one-step forecast $\hat{r}^{(A)}_{t+1}$, scored post hoc against realised returns by the out-of-sample R^2 (14) against a constant-mean benchmark, tracked on a trailing 1000-period window, and by full-sample mean squared forecast error (MSFE). With zero adoption the realised return and the demand-adjusted return $x_{t+1} = r_{t+1} - \mu D_t$ differ only by unpredictable null-order impact, so this subsection reports realised-return performance only; the two targets separate once adopters start trading in phase 4.

The rule recovers the signal it was built for. The mean rolling out-of-sample R^2 is small, positive for every window, and rising with window length: 0.013 for $w = 50$, 0.033 for $w = 100$, and 0.047 for $w = 250$. Figure 2 shows the time profile: the traces co-move because they score one path, longer windows sit almost everywhere higher, and the share of periods with positive R^2 rises with w. The $w = 250$ trace is positive at every evaluated period (minimum 0.020), $w = 100$ in 98.5 percent of periods, while $w = 50$ dips below zero in roughly 16 percent. These levels are consistent with the return law's ceiling: an oracle knowing φ exactly would attain a population R^2 of $\varphi^2 = 0.0625$, which the longer windows approach from below as estimation noise shrinks. The coefficient estimates tell the same story: the rolling $\hat{\varphi}$ fluctuates around the input value, with means 0.195, 0.217, and 0.229 and standard deviations 0.145, 0.103, and 0.063 for $w = 50$, 100, and 250; the mild downward gap reflects the small-sample bias of least-squares AR(1) estimation together with single-path sampling noise, and closes as the window lengthens. Full-sample MSFE is nearly flat across windows (1.18×10^{-4} to 1.15×10^{-4}) and the rolling MSFE traces show no trend, as expected when no trader acts on the forecast. Sign accuracy, a diagnostic rather than an endpoint, is 0.56 for the two shorter windows and 0.57 for $w = 250$.

This completes the pre-adoption baseline. The previous subsection showed that the market does not drift when no one trades on a forecast; this one shows that the rolling rule recovers the linear-persistence signal at a known level, a small, stable, positive out-of-sample R^2 due entirely to the φr_t term.

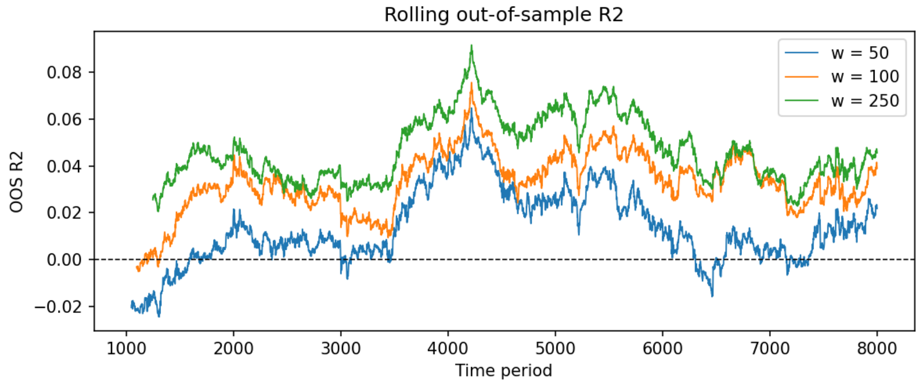

Figure 2: Rolling out-of-sample R^2 of the one-step AR(1) forecast against realised returns, computed on a trailing 1000-period window, for estimation windows $w = 50$, 100, and 250 applied to the same null-market path (input $\varphi = 0.25$, seed 43). The dashed line marks zero. The traces co-move because they score one path; longer windows sit almost everywhere higher (the $w = 100$ and $w = 250$ traces briefly cross around periods 6800 to 6900), the $w = 250$ trace stays positive at every evaluated period, and the $w = 50$ trace intermittently dips below zero.

6.3 Dual-channel result under stochastic adoption (phase 4)

Phase 4 turns the offline forecast of the previous subsection into a tradeable rule and lets its use spread. Adopters map the shared rolling AR(1) forecast ($w = 250$) into positions through the mean-variance demand (4), clipped at the position cap as in (5); non-adopters keep the random null rule. With combined risk scaling $a\sigma^2 = 0.001$ and per-trader cap $\bar{q} = 0.05$ the cap binds in nearly every period, so adopters are effectively fixed-size traders on the forecast's sign and their impact stays comparable in magnitude to the φr_t term rather than dominating it. Adoption follows the stochastic diffusion of (8) and (9) in three regimes: zero adoption ($\pi = 0$), slow diffusion ($\pi = 3 \times 10^{-4}$), and fast diffusion ($\pi = 10^{-3}$), each with exit probability $\delta = 0$, run on paired shocks from the shared seed 71 so that differences across regimes are attributable to adoption alone. The market is otherwise the phase 3 configuration ($N = 200$, $T = 8000$, $\mu = 0.05$, input $\varphi = 0.25$, $\sigma = 0.01$, $\sigma_q = 1.0$), with every rolling metric computed on a trailing 1000-period window. Adoption is held at zero until period 1250, the first period with a complete metric window, and the headline comparison contrasts a low-adoption window (periods 1250 to 2750) with a high-adoption window (the final 1500 periods), by which point the slow regime has reached a mean adoption share of 0.836 and the fast regime 0.994.

Against realised returns, the forecast improves as adoption rises. In the high-adoption window the fast regime's rolling out-of-sample R^2 against r_{t+1} is 0.190, versus 0.070 for the zero-adoption control on the same shocks (0.065 in the control's own late window), with the slow regime in between at 0.165. The control row is the right low-adoption anchor: diffusion at $\pi = 10^{-3}$ is fast enough that the fast regime's own early window already averages an adoption share of 0.505, with realised R^2 already lifted to 0.094. The mechanism is the decomposition $r_{t+1} = \mu D_t + x_{t+1}$ of (12): every adopter trades in the direction of the same public forecast, so coordinated adopter demand pushes the realised return in the forecast direction, and the forecast scores partly against its own price impact. The left panel of Figure 3 shows that this is not a one-seed accident: pooled across 100 Monte Carlo seeds, the mean realised-return R^2 rises monotonically in adoption share.

The same forecast on the same path is simultaneously losing the signal it was built to exploit. Scored against the demand-adjusted return $x_{t+1} = r_{t+1} - \mu D_t$, the counterfactual that strips out the contemporaneous demand impact while leaving the lagged-return component φr_t and the news shock in place (it is not the full regression residual; section 3.6), the rolling out-of-sample R^2 falls to 0.039 in the fast regime's high-adoption window, versus 0.082 under the control, with the slow regime at 0.047. This reading asks whether the forecast still carries independent predictive content once self-fulfilment is removed, and the answer weakens as adoption rises: in the right panel of Figure 3 the Monte Carlo mean declines as adoption rises, with the drop steepening at high adoption shares.

The rolling effective AR coefficient is the mechanical link between the two readings. Estimated as the lag-1 autocorrelation of realised returns on the same trailing window, it rises from 0.282 under the control, close to the input $\varphi = 0.25$, to 0.447 in the fast regime's high-adoption window (Figure 4). Adopter demand responds positively to recent returns through the forecast, so the impact term μD_t reinforces the lagged-return component and measured persistence grows with adoption. The rolling AR fit estimates its coefficient from realised returns, absorbs this amplified effective coefficient, and therefore systematically over-predicts the demand-adjusted process, whose own AR coefficient stays at the input 0.25 by construction; that over-prediction is the demand-adjusted decline of the previous paragraph. The control, meanwhile, moves far less across the same two windows (realised 0.070 to 0.065, demand-adjusted 0.082 to 0.068, effective coefficient 0.282 to 0.278), so the movements in the diffusion regimes are adoption effects, not time effects.

This is the dual-channel result, and it is the headline finding of the report: adoption raises realised-return predictive performance through the self-fulfilment channel while eroding the independent demand-adjusted signal, on the same path and from the same forecast. Which reading a forecaster sees depends entirely on whether the evaluation target includes the forecaster's own price impact.

6.4 Endogenous CE-based switching (phase 5)

Phase 5 makes adoption respond to how the forecast is actually paying: the stochastic diffusion of (8) and (9) is replaced by the risk-adjusted performance-based switching of section 3.4. Each period the simulator records the realised one-period profit $q_{i,t}\, r_{t+1}$ for a representative trader under each rule: the common capped adopter position, and a representative non-adopter's order. Each rule's certainty equivalent $\mathrm{CE}_t^{(k)} = \bar{\Pi}_t^{(k)} - \frac{a}{2}\hat{\sigma}_t^{2,(k)}$, the rolling mean profit net of a risk penalty on its rolling variance, is computed over $W_A = 500$ periods with $a = 1$, per (10), and each non-adopter switches to the advanced rule with the logistic probability $\Lambda(\alpha + \beta S_t)$ of (11) in the score gap $S_t = \mathrm{CE}_t^{(A)} - \mathrm{CE}_t^{(0)}$, with $\alpha = -5$ and $\beta = 10000$. At a zero score the implied switch probability is $\Lambda(-5) \approx 0.0067$ per period, and score movements of order 10^{-4} move it strongly in either direction. Everything else stays at the phase 4 configuration with adoption starting at period 1250; the comparator is the fast stochastic regime ($\pi = 10^{-3}$) rerun on paired

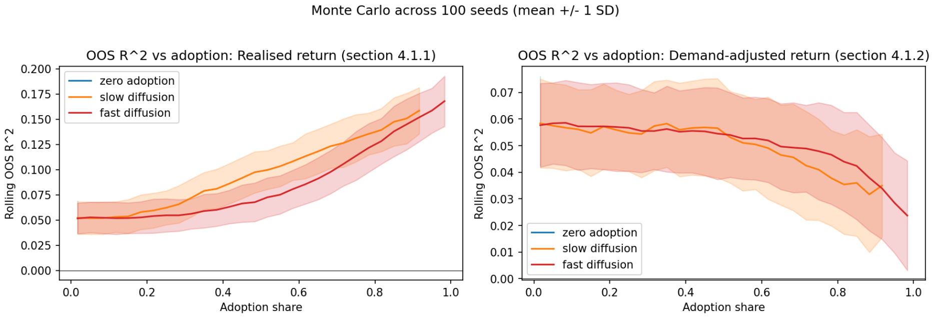

Figure 3: Rolling out-of-sample R^2 of the shared AR(1) forecast versus adoption share, pooled across 100 Monte Carlo seeds. The left panel scores the forecast against the realised return r_{t+1} (self-fulfilling channel); the right panel scores the same forecast against the demand-adjusted return $x_{t+1} = r_{t+1} - \mu D_t$ (demand-adjusted-erosion channel); note the different y-axis scales. Lines are means within 30 adoption-share bins, pooling post-warm-up periods across seeds; shaded bands are ± 1 standard deviation; bins with 10 or fewer observations are dropped, which trims the slow regime's curves short of full adoption. The zero-adoption regime contributes observations only at adoption share zero, a single bin, so it appears in the legend but draws no visible curve. The mean realised-return R^2 rises from about 0.05 near zero adoption to about 0.17 at full adoption, while the mean demand-adjusted R^2 falls from about 0.06 to about 0.02. The slow-diffusion curves sit slightly to the left of the fast curves because the trailing 1000-period metrics lag the instantaneous adoption share on the x axis, and the lag matters more the faster adoption ramps.

shocks from the shared seed 71, so any difference is attributable to the adoption mechanism.

The adoption path changes shape; the erosion does not. The endogenous share is steplike. Through most of the run-up the score hovers slightly below zero (negative in 88 percent of pre-saturation periods, held down in part by the null rule's positive own-impact profit mean $\mu\sigma_q^2/N = 2.5 \times 10^{-4}$), and at the median pre-saturation score the implied switch rate is 0.00049 per period, about half the comparator's diffusion rate, so the share mostly creeps; episodes of positive score raise the per-period inflow by a factor of about 3.6, producing the bursts. The endogenous run reaches full adoption by period 2842, while the stochastic share crosses 0.99 only at period 6164. Yet over the same low- and high-adoption windows as the previous subsection the two regimes give nearly identical erosion readings: the demand-adjusted out-of-sample R^2 falls from 0.071 to 0.039 under stochastic diffusion and from 0.076 to 0.037 under performance switching, while the rolling effective AR coefficient rises from 0.320 to 0.447 and from 0.315 to 0.449 respectively, at mean adoption shares of 0.505 to 0.994 and 0.455 to 1.000. Figure 5 traces both diagnostics against the adoption share itself: the two regimes start from the same pre-adoption values, decline and rise along overlapping paths, and converge at full adoption. The performance-switching curve sits somewhat above the stochastic one on the demand-adjusted panel at intermediate shares; this is a lag artefact: the endogenous run sweeps those shares too quickly for the trailing 1000-period metric window to refresh (235 periods between shares 0.6 and 0.95, against 1945 for the diffusion run).

Endogenous switching therefore changes when the market visits each adoption level, not what adoption does once there: at matched shares both mechanisms produce the same demand-adjusted decline and the same effective-coefficient amplification. The dual-channel result of the previous subsection is thus not an artefact of the exogenous diffusion assumption; erosion is a property of the adoption share, however the share is generated. The run also shows why

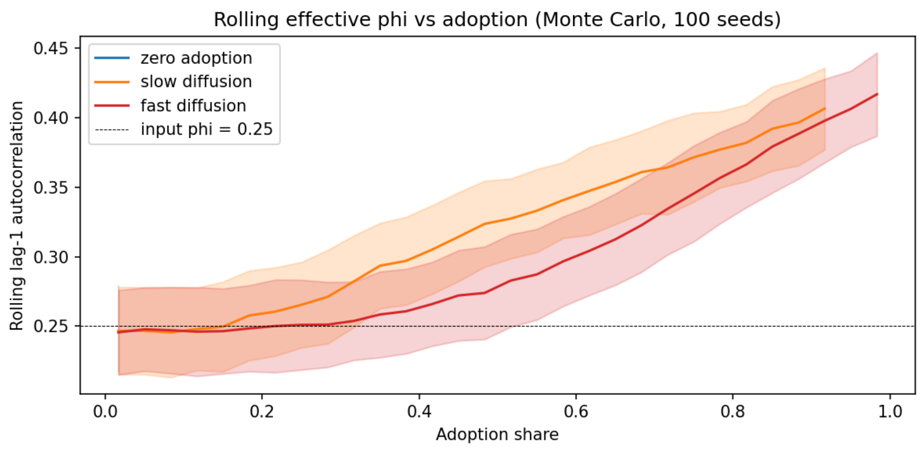

Figure 4: Rolling lag-1 autocorrelation of realised returns versus adoption share, same Monte Carlo design as Figure 3 (100 seeds, 30 adoption-share bins, mean ± 1 standard deviation, bins with 10 or fewer observations dropped). The dashed line marks the input $\varphi = 0.25$. Both diffusion regimes start close to the input value near zero adoption and rise steadily with adoption share, reaching roughly 0.40 to 0.42 at their highest observed adoption shares; the zero-adoption regime occupies a single bin and draws no visible curve.

the mechanism reinforces rather than brakes adoption in the cost-free baseline: adopter trades strengthen the autoregressive component of realised returns, so the advanced rule's certainty equivalent pulls level and then ahead as adoption builds (the score's pre-saturation maximum arrives one period before full adoption), and nothing pushes the score decisively negative at high adoption. Whether transaction costs change that is taken up in section 6.6.

6.5 Parameter sweep and critical adoption shares (phase 6)

Phase 6 turns the single-configuration findings of phases 4 and 5 into a map. The sweep crosses $\mu \in \{0.025, 0.05, 0.075, 0.10, 0.15\}$, $\varphi \in \{0.05, \ldots, 0.30\}$, forecast windows $w \in \{100, 250, 500\}$, and AR orders $p \in \{1, 2, 5, 10\}$, with AR(1) the baseline rule of section 3.3 and the higher orders a robustness extension, at 10 seeds per cell (3600 runs). Everything else follows the phase 4 design ($N = 200$, $T = 8000$, trailing 1000-period metrics) under slow stochastic diffusion ($\pi = 3 \times 10^{-4}$, $\delta = 0$). Adoption stays at zero until a 1500-period baseline window has completed, so each run measures its own pre-adoption baseline; by the end of the run the share reaches roughly 0.8. Seeds are paired across p: the same $(w, \varphi, \mu, \text{seed})$ gives the same shocks and adoption path at every order, so differences across p isolate the forecast specification.

The zero-crossing threshold $A^*_{R^2,\text{da}}$ of section 4.4 fires only where price impact is strong or the baseline signal is near zero; at the phase 4 and 5 settings the demand-adjusted R^2 falls but stays positive. Phase 6 therefore reports the relative analogue, fixed before the sweep was run and defined wherever the low-adoption baseline is positive. $A^*_{R^2,\text{da,rel}}$ is the smallest adoption share, past the baseline window, at which the rolling demand-adjusted R^2 has fallen to half its own low-adoption baseline (threshold factor 0.5); the same convention gives $A^*_{R^2,\text{realised,rel}}$ on the realised target and $A^*_{\varphi,\text{rel}}$ on the effective AR coefficient, the smallest share at which the rising effective coefficient reaches 1.5 times its baseline. A cell reports the mean crossing share over its crossing seeds, and only when at least half of the 10 seeds cross, so a noisy minority cannot

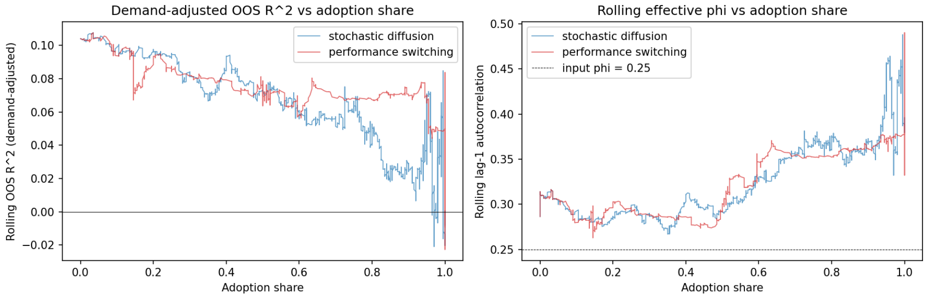

Figure 5: Rolling out-of-sample R^2 against the demand-adjusted return (left) and rolling lag-1 autocorrelation of realised returns (right), plotted against the adoption share for exogenous stochastic diffusion ($\pi = 10^{-3}$) and endogenous CE-based switching on the same paired shocks (seed 71); both metrics use a trailing 1000-period window. The two regimes share the pre-adoption segment (demand-adjusted R^2 near 0.104 and effective coefficient between 0.286 and 0.314 at zero adoption, against the input $\varphi = 0.25$ marked by the dashed line), then decline (left) and rise (right) along overlapping paths as the share grows, converging at full adoption. The performance-switching curve sits above the stochastic one on the left panel at intermediate shares because the endogenous run sweeps those shares in few periods relative to the metric window; the vertical segments at share 1.0 collect each run's post-saturation stretch.

set it. One caveat applies throughout: the rolling metrics trail adoption by up to the 1000-period evaluation window, so each A^* is a detection point, an upper bound on the erosion onset, not the onset itself.

Figure 6 maps the demand-adjusted threshold for the AR(1) rule: it is reached in 67 of the 90 cells, at a mean detected share of 0.29. Two regularities organise the map. The crossing arrives later as the input φ grows, from very low shares in the $\varphi = 0.10$ row to 0.37 to 0.65 at $\varphi = 0.30$: the threshold scales with a baseline that grows with φ, so stronger signals need more adoption to lose half their content. And the crossing is roughly flat in μ (column means 0.25 to 0.31, no monotone trend), because the baseline R^2 also grows with μ and the relative threshold scales with it. The μ gradient lives instead in the depth of the erosion: the mean demand-adjusted R^2 over the final 1500 periods falls with μ in every φ row, from $+0.08$ at $(\varphi = 0.30, \mu = 0.025)$ to -0.28 at $(\varphi = 0.05, \mu = 0.15)$ for $w = 250$, so at the strongest price impact the independent signal is not merely halved but pushed below zero, the absolute erosion criterion of section 4.4. The effective-coefficient threshold (Figure 7) is reached in 82 of 90 cells at a mean share of 0.38, and there the μ gradient shows in the threshold itself: at moderate-to-high φ, larger μ pulls $A^*_{\varphi,\mathrm{rel}}$ earlier ($w = 250$, $\varphi = 0.25$: 0.66 at $\mu = 0.05$, 0.33 at $\mu = 0.15$). The eight unreached cells sit at low μ and $\varphi \geq 0.20$, where feedback is too weak to amplify an already strong coefficient by half.

The realised-return threshold is the designed non-result. $A^*_{R^2,\mathrm{realised,rel}}$ is reached in only 28 of the 90 AR(1) cells, always early (mean share 0.06, maximum 0.21), with 26 of the 28 hits at $\varphi \leq 0.20$ and a median baseline realised R^2 of 0.014 at the hit cells. These are noise crossings, not erosion: where the baseline realised R^2 is near zero, the half-baseline threshold sits at essentially zero and rolling noise can dip below it without any systematic decline. The substantive realised-channel finding is the absence of crossings elsewhere: the tail-window realised R^2 rises with both φ and μ, up to 0.32 at the strongest cell, the self-fulfilment channel of phase 4 at grid scale. The same caveat covers the bottom rows of Figure 6: at $\varphi \leq 0.10$ the demand-adjusted baseline is also near zero, and those rows fail the filter or fire at near-zero shares.

The dual-channel pattern survives every AR order tested. At each p the effective-coefficient threshold is reached on 80 to 82 of 90 cells at mean shares 0.38 to 0.41, the tail demand-adjusted R^2 deepens with μ, and the realised threshold stays sparse (15 to 28 cells) and confined to low shares. Across the full four-order grid the demand-adjusted threshold is reached in 196 of 360 cells (54 percent, per-seed hit rate 0.53) at a mean share of 0.27, against a realised-channel hit rate of 0.26. Reachability falls with order (67, 59, 42, 28 of 90 cells for $p = 1, 2, 5, 10$) because the baseline falls with it (grid means 0.035, 0.028, 0.004, -0.041): a higher-order rule fits more coefficients on the same window and starts with less out-of-sample signal, so more cells fail the positive-baseline requirement or the hit-rate filter. Where both AR(1) and AR(p) set a value, the delta-from-AR(1) map (Appendix B) is mildly negative on average (-0.04 at $p = 2$, -0.14 at $p = 5$, -0.20 at $p = 10$): higher-order rules cross the relative threshold somewhat earlier, consistent with having less signal to lose, and nowhere does the mechanism reverse. The erosion of the independent signal is therefore a property of the mechanism, not of the AR(1) specification.

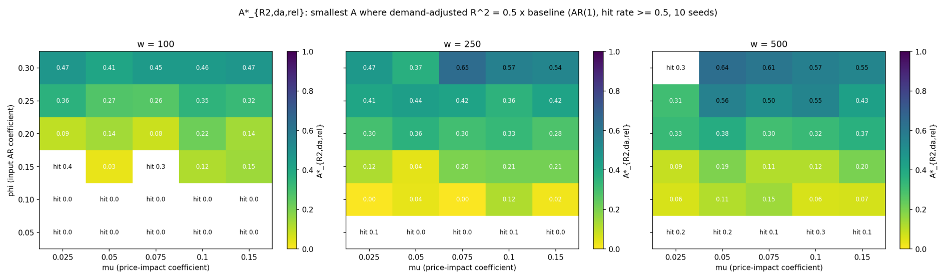

Figure 6: Relative demand-adjusted threshold $A^*_{R^2,\mathrm{da,rel}}$ across the (μ, φ) grid for the AR(1) rule, one panel per forecast window $w \in \{100, 250, 500\}$. Each cell shows the smallest adoption share at which the rolling demand-adjusted R^2 fell to half its low-adoption baseline, averaged over the crossing seeds; cells where fewer than half of the 10 seeds crossed print their hit rate instead (for example "hit 0.3") and are left blank. The crossing share rises with the input φ, from very low values in the $\varphi = 0.10$ row, where the baseline is itself near zero, to 0.37 to 0.65 at $\varphi = 0.30$, and is roughly flat across μ; the $\varphi = 0.05$ row is unreached in every panel.

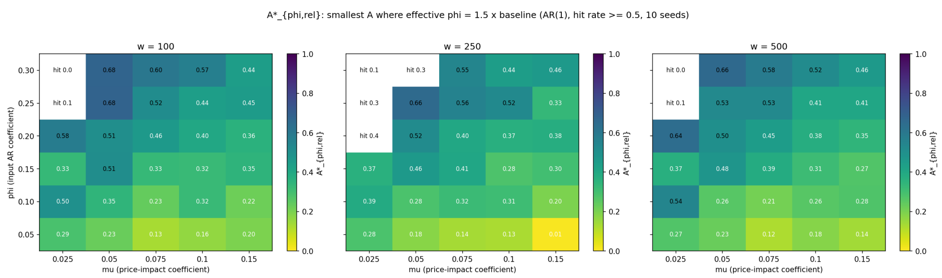

Figure 7: Relative effective-coefficient threshold $A^*_{\varphi,\mathrm{rel}}$, same layout and hit-rate convention as Figure 6: the smallest adoption share at which the rolling lag-1 autocorrelation of realised returns has grown to 1.5 times its low-adoption baseline. At moderate-to-high φ the threshold falls as μ grows (at $w = 250$, $\varphi = 0.25$: 0.66 at $\mu = 0.05$ down to 0.33 at $\mu = 0.15$), and the unreached cells cluster at low μ and $\varphi \geq 0.20$, where price impact is too weak to amplify an already strong baseline coefficient by half. The lowest φ row crosses earliest because a 50 percent relative increase on a small baseline is a small absolute move.

6.6 Transaction-cost extension and economic endpoint (phase 7)

Phase 7 closes the results with the transaction-cost extension and the economic endpoint it enables. A proportional cost c is charged on every unit traded, so per-period adopter net profit is $q_t^{(A)} r_{t+1} - c\,|q_t^{(A)}|$; the dynamics are unchanged and costs enter only the profit accounting, so 50 paired-shock seed runs of the phase 4 slow-diffusion design support the whole sweep over $c \in \{0, 0.001, 0.002, 0.003, 0.004, 0.005, 0.0075, 0.01\}$. Net profit is pooled across seeds and time into 30 adoption-share bins; each cost level reports the A^*_{profit} of section 4.4 in both conventions: $A^*_{\text{profit,abs}}$, the smallest share at which mean absolute net profit crosses zero, and $A^*_{\text{profit,rel}}$, the smallest share at which mean net profit relative to the null benchmark crosses zero. The benchmark, the null leg of the primary economic endpoint (16), is the representative null trader's realised gross profit minus its own cost bill $c\,E|q^{(0)}|$, with $E|q^{(0)}| = \sqrt{2/\pi}\,\sigma_q \approx 0.80$.

At zero cost the rule pays at every observed share and pays more as adoption grows, from a pooled mean of 1.2×10^{-4} per period at $A < 0.2$ to 2.0×10^{-4} at $A > 0.6$. That growth, roughly 0.8×10^{-4} between the two windows, recurs at every cost level: cost shifts the absolute curves down in parallel and leaves the slope in adoption untouched (Figure 8, left). The absolute threshold therefore exists only in a narrow band: the binned curve is positive at every observed share for $c \leq 0.002$, negative everywhere for $c \geq 0.005$, and crosses zero only at $c = 0.003$ and 0.004, at $A^*_{\text{profit,abs}} = 0.35$ and 0.72 (Figure 9). The crossing runs from below, negative at low adoption and positive at high: a profitability onset, the share self-fulfilment needs before the rule covers its costs, the reverse of the from-above erosion reading the threshold was designed to detect. The null-relative curve produces no meaningful crossing at all. At $c = 0$ it is slightly negative in 26 of the 28 populated bins and at both pooled ends (-2.0×10^{-4} at $A < 0.2$, -0.5×10^{-4} at $A > 0.6$); at every positive cost level it is uniformly positive. The single detected crossing, $A^*_{\text{profit,rel}} \approx 0.68$ at $c = 0$, is one bin popping to $+1.7 \times 10^{-4}$ against an in-bin standard deviation of 0.011: noise around a systematically negative mean, not a threshold.

Both null-relative readings are statements about position size rather than forecast content. A null trader's own order moves the within-period quote, so its expected gross profit is not zero but $\mu\sigma_q^2/N = 2.5 \times 10^{-4}$ per period of own-impact paper profit, booked on a mean absolute position $E|q^{(0)}| \approx 0.80$, roughly 16 times the adopter cap $\bar{q} = 0.05$. At $c = 0$ that term exceeds adopter gross profit at every observed share, so the null-relative mean is systematically, if slightly, negative throughout; only the deficit's narrowing with adoption is real. At $c > 0$ the same asymmetry flips sign through the cost bill: the null pays cost on $|q^{(0)}| \approx 0.80$ while the adopter pays on at most 0.05, so the curve sits uniformly above zero at a level set by the cost difference rather than by the forecast. All profits, null and adopter alike, are marked at the post-impact quote with no unwind flow, so every reading here is a paper profit under the model's own accounting. The null-relative endpoint is therefore implemented exactly as designed but dominated by position-size asymmetry in both directions, and the absolute endpoint is the cleaner economic read in this baseline. On that read adopter mean profit grows with adoption at every cost level tested; transaction costs set only the level at which the growth starts to pay.

7. Discussion

7.1 Interpretation of the dual-channel mechanism

7.2 Why the effective AR coefficient rises with adoption

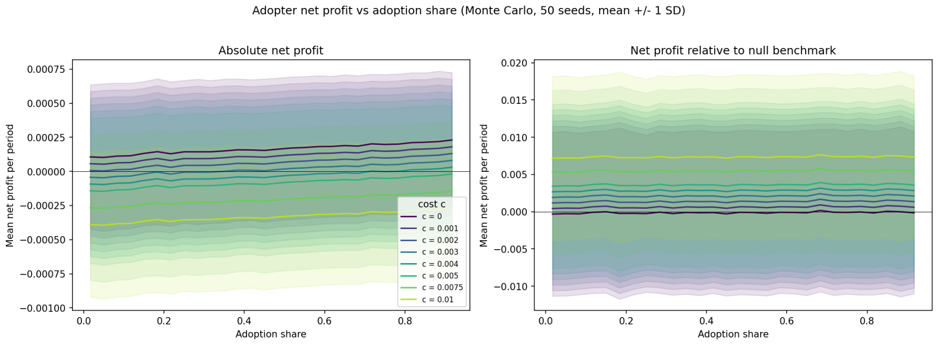

Figure 8: Per-period adopter net profit versus adoption share at the eight cost levels, pooled over 50 paired-shock seeds into 30 adoption-share bins (lines: bin means; bands: ± 1 standard deviation of the in-bin observations; bins with 10 or fewer observations are dropped, which ends the curves short of full adoption). Left: absolute net profit $q_t^{(A)} r_{t+1} - c|q_t^{(A)}|$. The curves rise with adoption at every cost level and rising cost shifts them down in parallel, so a zero crossing exists only in the $c = 0.003$ to 0.004 band. Right: net profit relative to the null benchmark, on a vertical scale more than an order of magnitude larger that is set by the null's cost bill: the $c = 0$ curve sits slightly below zero throughout while every positive-cost curve sits uniformly above it, ordered by cost level.

7.3 Boundary conditions and the realised-channel non-result

7.4 Limitations

8. Conclusion

8.1 Summary of findings

8.2 Implications for forecasting under adoption

8.3 Directions for future work

References

A. Beja and M. B. Goldman. On the dynamic behavior of prices in disequilibrium. *Journal of Finance*, 35(2):235–248, 1980.

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J. D. Farmer and S. Joshi. The price dynamics of common trading strategies. *Journal of Economic Behavior & Organization*, 49(2):149–171, 2002.

A. Model equations and derivations

B. Additional figures

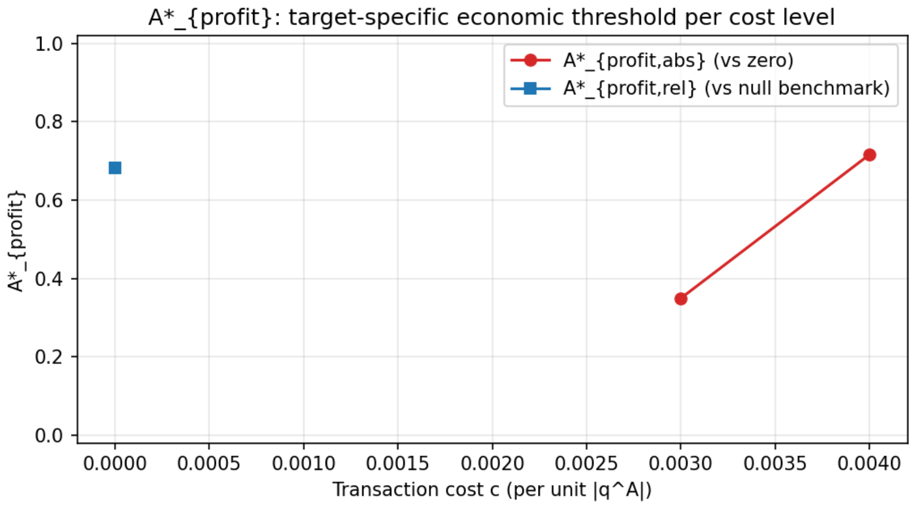

Figure 9: Per-cost critical adoption shares A^*_{profit}. Red circles: $A^*_{\text{profit,abs}}$, the smallest binned share at which mean absolute net profit crosses zero from below; blue square: $A^*_{\text{profit,rel}}$, the same crossing for the null-relative curve. A marker appears only where a crossing exists, so the horizontal axis spans only the cost levels that produce one: the absolute threshold is defined at $c = 0.003$ ($A^* = 0.35$) and $c = 0.004$ ($A^* = 0.72$) and nowhere else on the grid, absolute net profit being positive at every observed share for $c \leq 0.002$ and negative at every observed share for $c \geq 0.005$, while the null-relative threshold fires only at $c = 0$ ($A \approx 0.68$), the bin-noise crossing discussed in the text.

C. Parameter tables and reproducibility details